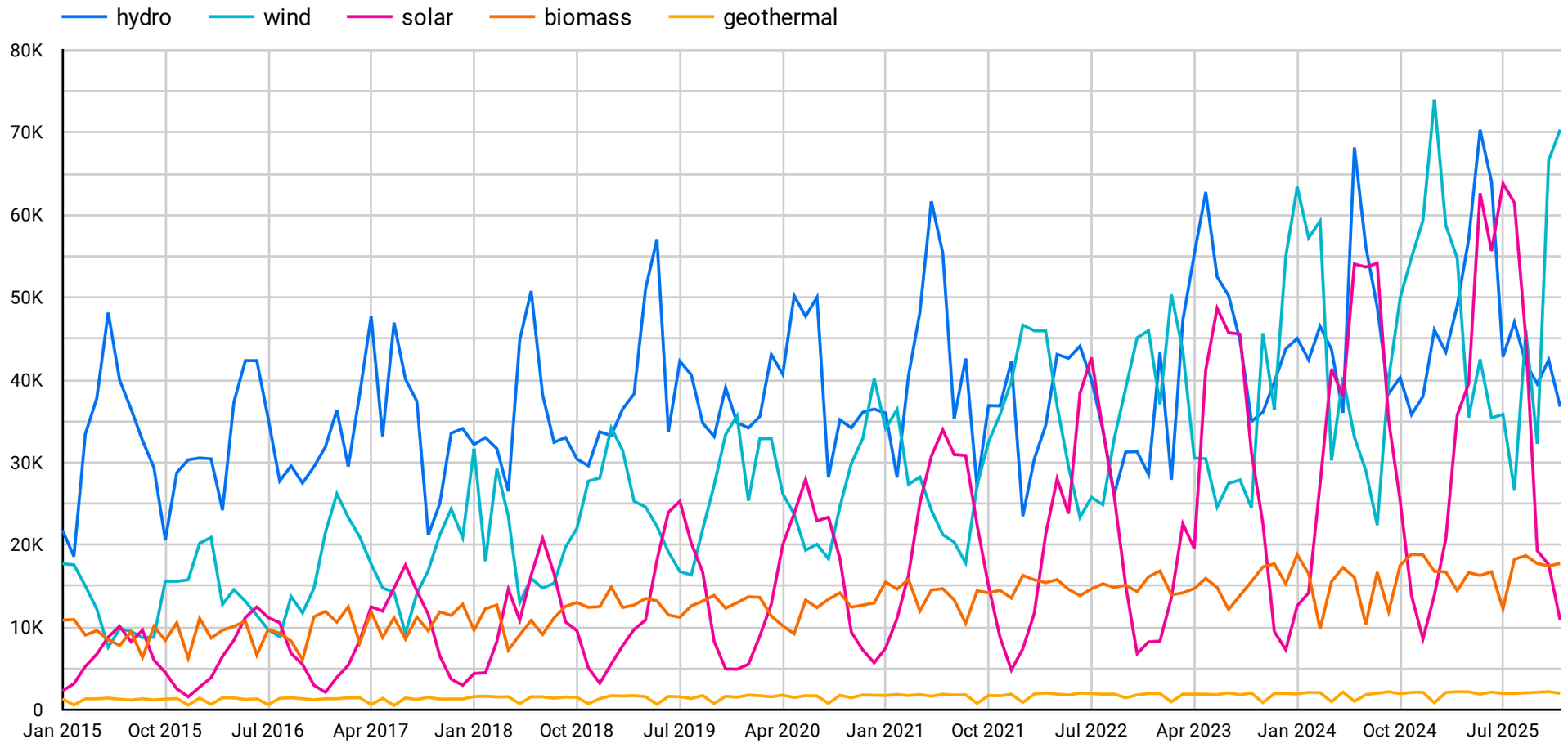
Renewable energy trends | Europe, 2015–2025

Select date range

Recorded generation (GWh)

13.2M

Monthly generation by renewable source (GWh)

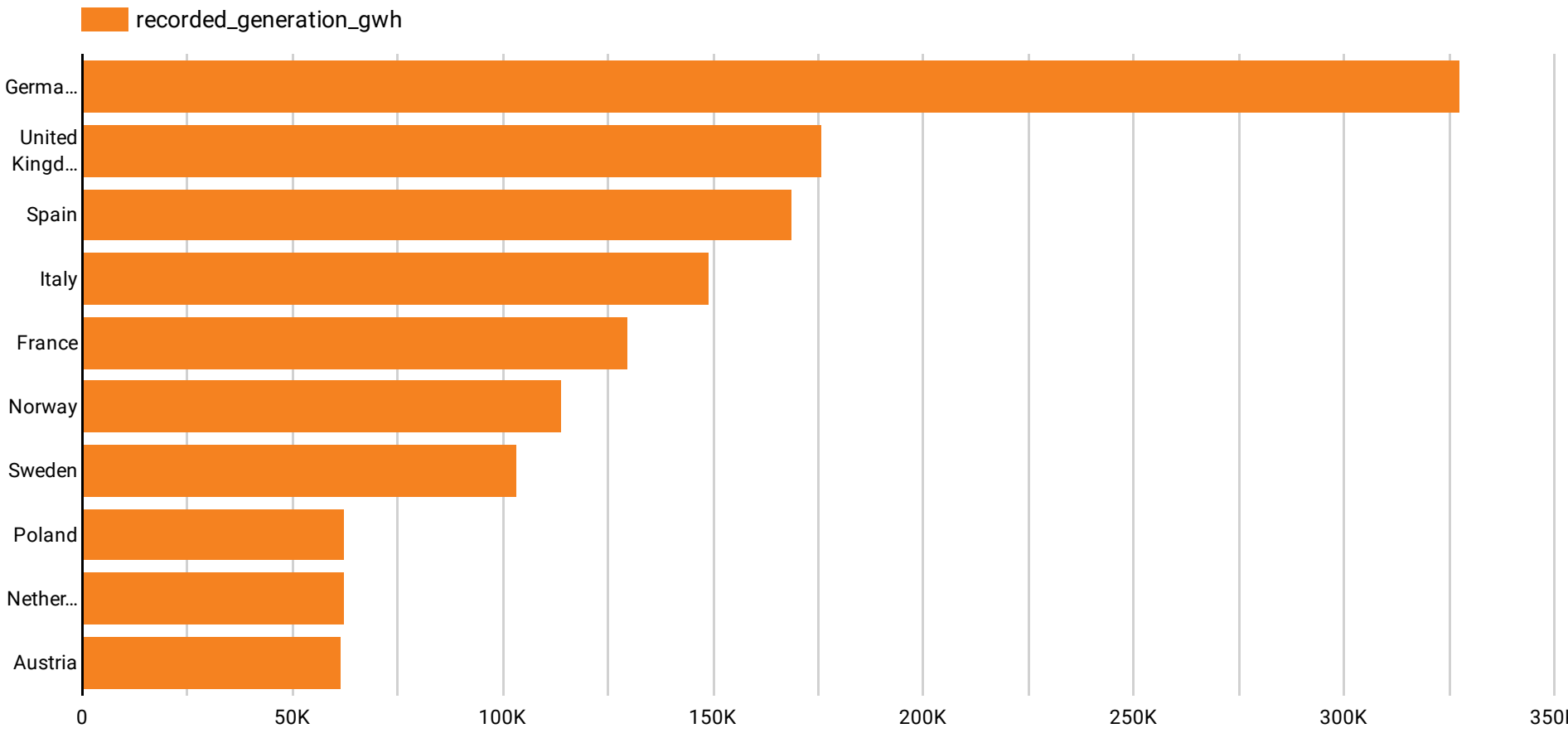


country_name

02 Country comparisons | Recorded renewable generation

Select date range

Recorded renewable generation (GWh) by country



year: 2025 (1)

Country detail — GWh and MWh per person (selected year)

	country_name	recorded_generation_gwh	recorded_generation_mwh...
1.	Germany	327,513.79	3.94
2.	United Kingdom	175,713.6	2.61
3.	Spain	168,774.5	3.56
4.	Italy	148,922.05	2.52
5.	France	129,577.53	1.91
6.	Norway	113,777.33	21.07
7.	Sweden	103,214.2	9.83
8.	Poland	62,416.09	1.66
9.	Netherlands	62,240.76	3.54
10.	Austria	61,496.5	6.83

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03 2030 targets — supplied values (%) | 26 countries have targets; 8 are unavailable. Targets are unverified metadata. Consumption varies across source rows, so a defensible current share or target gap cannot be calculated.

Select date range

Supplied 2030 renewable-share targets (%)

	Country	2030 target (%) ▾
1.	Poland	80
2.	Finland	80
3.	Greece	80
4.	Belgium	80
5.	Ireland	65
6.	Slovenia	65
7.	Malta	65
8.	Italy	65
9.	Hungary	60
10.	Slovakia	60
11.	Portugal	60
12.	Estonia	55
13.	Germany	55
14.	Sweden	50
15.	Romania	45
16.	Czechia	45
17.	Bulgaria	42.5
18.	Cyprus	40
19.	Spain	40
20.	Austria	40
21.	Netherlands	40
22.	France	40
23.	Serbia	32

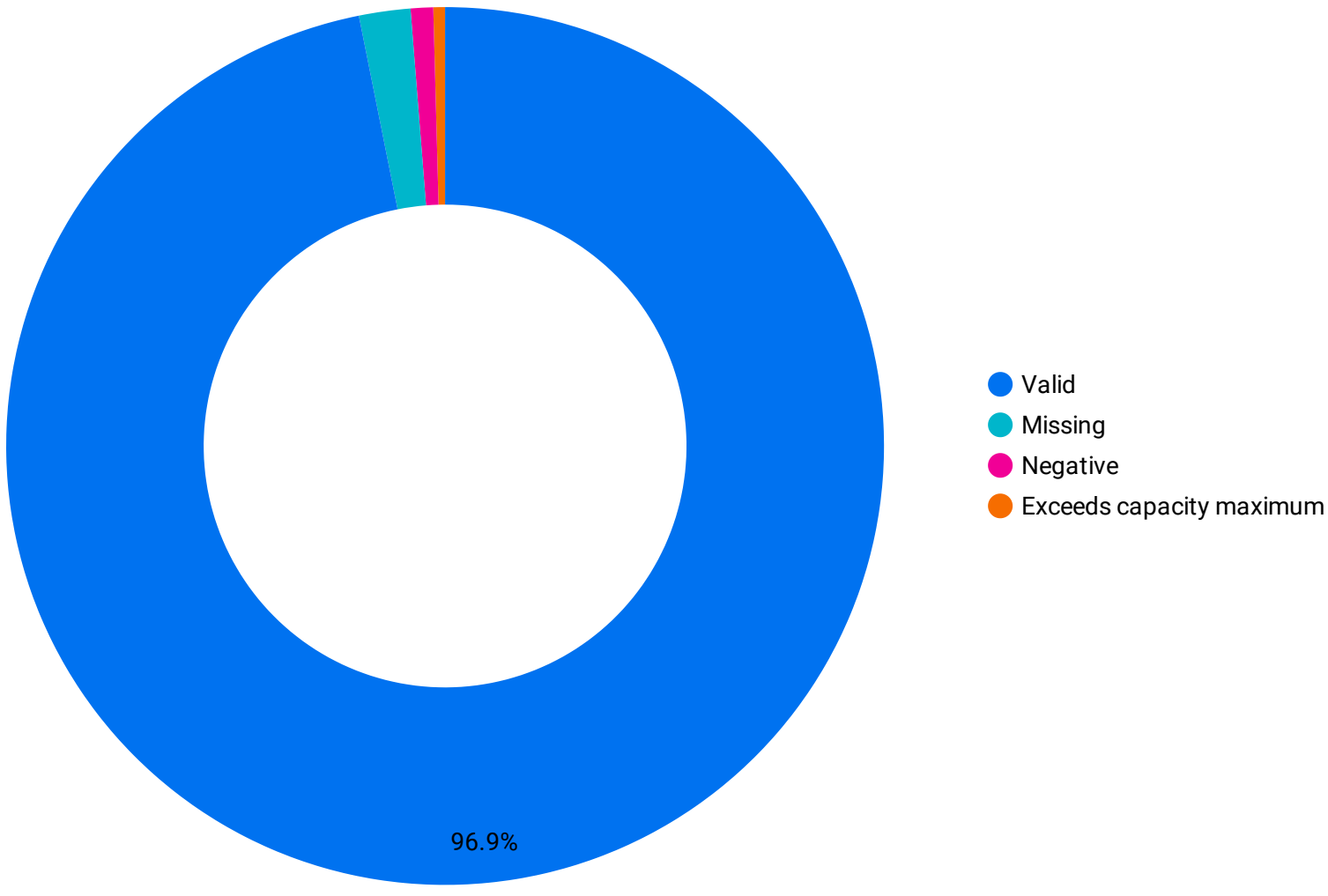
1 - 34 / 34

country_name

04 Data quality | 17,907 records; 96.85% valid. Missing: 339; negative: 148; exceeds capacity: 77. Invalid generation is excluded. Provider labels and targets are supplied metadata; provenance is unverified.

Select date range

Generation record quality — all years



country_name

Record counts by quality — selected country and dates

	quality_label	Record Count
1.	Valid	17,343
2.	Missing	339
3.	Negative	148
4.	Exceeds capacity maximum	77

1 - 4 / 4

METHOD & LIMITATIONS Generation = SUM(valid_generation_gwh). Invalid records stay blank and are excluded; coverage is incomplete. Quality rate = SUM(is_valid_generation) / SUM(record_count). Full dataset: 17,343 / 17,907 = 96.85%. Annual and monthly generation reconcile within 0.000001 GWh. No duplicate record IDs or country-month-source keys. Coverage: 29 countries in 2015, 30 in 2016, 32 in 2017, 34 from 2018. Changes in coverage affect trend comparisons. Switzerland 2018 and Malta 2019 have 11 recorded months. Twelve months does not mean all sources are complete. Per person = GWh / (population_millions × 1,000), in MWh/person. Supplied population is constant across years. 2030 targets use MAX by country, never SUM. 26 countries have targets; 8 are missing. These are supplied, unverified values. Consumption differs within 4,311 of 4,354 country-month groups. Do not sum or average source-level renewable shares or calculate a target gap. Estimated and provisional records remain included when generation is valid. No forecast, official target validation or causal claim is made. Sources: the two supplied CSV files (2015–2025). Portfolio analysis by Noah Nentwig.